D: A good starting point for the fitting procedure

According to eqn. (3.2), the theoretical expression for the velocity v_n after n revolutions can be written as

$$v_n = (v_0 + w)\, e^{-\beta s_n} - w\,;\ w = \frac{\alpha}{\beta}\ , \qquad \text{(D1)}$$

where $s_n = n \cdot 2\pi R$ is the distance driven after n revolutions. To determine the values of the fitting parameters v_0, α and β which minimize the least square sum

$$\varepsilon = \sum_{n=0}^{N} (v_n - u_n)^2 \qquad \text{(D2)}$$

where $u_0, u_1, u_2, ..., u_N$ are the measured velocities, the computer asks for some fairly good values to start with. These can be obtained by repeated guessing. It is, however, much easier to use the following formula, which usually will give a *very* good fit.

Let $m \le N$ be a number so that u_m and u_{2m} both are among the measured $u_0, u_1, u_2, ..., u_N$. We can then obtain the three *exact* agreements

$$v_0 = u_0\ ,\ v_m = u_m \text{ and } v_{2m} = u_{2m} \qquad \text{(D3)}$$

by choosing $v_0 = u_0$,

$$\alpha = w \cdot \beta;\quad w = \frac{u_m^2 - u_0 u_{2m}}{u_0 + u_{2m} - 2u_m}\ ,$$

$$\beta = \frac{1}{s_m} \ln\left(\frac{u_0 - u_m}{u_m - u_{2m}}\right) \qquad \text{(D4)}$$

To verify this assertion, we should solve the two equations

$$u_m + w = (u_0 + w)\, e^{-\beta s_m}\ , \qquad \text{(D5a)}$$

$$u_{2m} + w = (u_0 + w)\, e^{-\beta s_{2m}}\ , \qquad \text{(D5b)}$$

for w and β. This can be done by noting the identity between the two fractions,

$$\frac{u_m + w}{u_{2m} + w} = \frac{(u_0 + w)\, e^{-\beta \cdot s_m}}{(u_0 + w)\, e^{-\beta \cdot s_{2m}}} = e^{\beta (s_{2m} - s_m)} = e^{\beta \cdot s_m} \text{ and} \qquad \text{(D6a)}$$

$$\frac{u_0 + w}{u_m + w} = \frac{(u_0 + w)}{(u_0 + w)\, e^{-\beta \cdot s_m}} = e^{\beta \cdot s_m}\ , \qquad \text{(D6b)}$$

which implies that

$$(u_m + w)^2 = (u_0 + w)(u_{2m} + w)\,;\ \text{i.e.}$$

$$u_m^2 + 2\,w u_m + w^2 = v_0\, u_{2m} + u_0 w + w u_{2m} + w^2 \qquad \text{(D7)}$$

With w^2 being eliminated, isolation of w gives the result postulated in (D4). Inserting this, the left hand side of (D6a) can be written as

$$\frac{u_m+w}{u_{2m}+w} = \frac{u_m(u_0+u_{2m}-2u_m)+(u_m^2-u_0u_{2m})}{u_{2m}(u_0+u_{2m}-2u_m)+(u_m^2-u_0u_{2m})}$$
$$= \frac{u_mu_0+u_mu_{2m}-u_m^2-u_0u_{2m}}{u_{2m}^2-2u_{2m}u_m+u_m^2}$$
$$= \frac{u_0(u_m-u_{2m})-u_m(u_m-u_{2m})}{(u_m-u_{2m})^2}$$
$$= \frac{(u_0-u_m)(u_m-u_{2m})}{(u_m-u_{2m})^2} = \frac{u_0-u_{2m}}{u_m-u_{2m}} \qquad \text{(D8)}$$

Inseting this in (D6a) and solving for β , we obtain the result postuled in (D4).

Let us check that use of (D6b) gives the same result: Here insertion of w from (D4) gives

$$\frac{u_0+w}{u_m+w} = \frac{u_0(u_0+u_{2m}-2u_m)+(u_m^2-u_0u_{2m})}{u_m(u_0+u_{2m}-2u_m)+(u_m^2-u_0u_{2m})}$$
$$= \frac{u_0^2-2u_0u_m+u_m^2}{u_mu_0+u_mu_{2m}-u_m^2-u_0u_{2m}}$$
$$= \frac{(u_0-u_m)^2}{u_0(u_m-u_{2m})-u_m(u_m-u_{2m})}$$
$$= \frac{(u_0-u_m)^2}{(u_0-u_m)(u_m-u_{2m})} = \frac{u_0-u_{2m}}{u_m-u_{2m}} \qquad \text{(D9)}$$

in full accordance with (D8).